S2 E11

BK

Season 2: Retrieval and RAG

Field note

The citation rule.

E9 and E10 each added a rule to the prompt. This field note puts both to a full test: does requiring a citation change what the model says? No new runs, no cost.

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Field note

This episode is a field note

No new idea this time. A field note takes lessons already taught and puts them to a full test.

LESSON FROM E9

Allow "not found"

One sentence raised correct "not found" replies from 83 to 105 of 120.

LESSON FROM E10

Require citations

Every answer cited a passage, and code checked each citation.

The open question

E10 added citations on top of E9’s sentence, so its results mixed the two. Do the lessons hold together, and did citing change the answers themselves?

HOW TO READ IT

Question → Setup → Result → Limits → Decision rule

Why it matters

The worry: citations might change the answers

An instruction that asks for more can change more than you meant. Three ways the citation rule could have done that:

More cautious?	Having to point to a source, the model might say "not found" more often, even when the answer is there.
Different answers?	It might pick different facts, the ones easiest to cite, and get some questions wrong that it used to get right.
Longer answers?	Every claim needs a marker. Answers might grow, and every extra token is paid for.

This field note

Does requiring a citation change what the model says, or only how it says it?

The method

No new runs: compare two that already exist

E9 and E10 each ran the full system on all 520 test questions. Their instructions differ by one idea.

NOT CITED: FROM E9

```
Answer the question using the passages
below. If the passages do not contain the
answer, reply that you could not find it.
```

CITED: FROM E10

```
Answer the question using only the passages
below. After each claim, cite the passage
it comes from, as [P1], [P2] and so on. If
the passages do not contain the answer,
reply that you could not find it.
```

Same everything else	Same model, same judge, same 520 questions. Every prompt held the same five passages in both runs, checked for all 520.

Why this works

Only the instruction differs, so any change in the answers comes from it. A controlled comparison, for free.

Three questions, checked question by question

Totals can hide movement. So each of the 520 questions is compared across the two runs.

Right answers	Of the 400 answerable: did citing change which ones were right?
Said not found	Of the 120 the library cannot answer: did citing change when the model said so?
Length	Did citing make answers longer? Counted in output tokens, the part you pay most for.

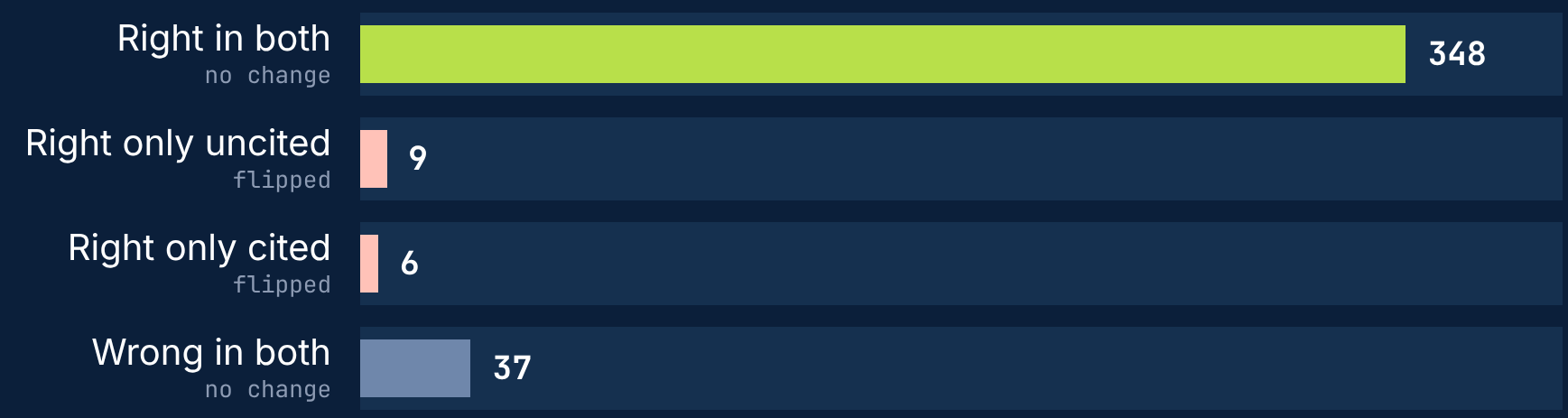
How to read a flip

A flip is a question right in one run and wrong in the other. Flips both ways in similar numbers mean noise. Flips mostly one way mean a real effect.

Finding

Citing did not change which answers were right

Right answers: 357 of 400 without citations, 354 with them. Here is each of the 400, compared across the two runs.



The 15 flips

9 one way, 6 the other. A split like that happens by chance about 6 times in 10.

So

385 of the 400 questions got the same verdict in both runs. Citing did not change what the model knew to say.

Finding

Citing did not change when it said “not found”

The 120 questions the library cannot answer, compared one by one.

NOT CITED

105 of 120

said "not found"

CITED

107 of 120

said "not found"

Question by question	101 said "not found" in both runs. 6 more did only when citing, 4 only without.
So	No sign of a more cautious model. The jump to 105 came from E9’s "not found" sentence, which both runs share.

Finding

Citations added two tokens, not sentences

Median answer length, in output tokens, across all 520 answers: 26 without citations, 28 with them.

THE E-2194 QUESTION, IN BOTH RUNS

What does error E-2194 mean?

Not cited	"E-2194 means the Ashby tumble dryer's factory reset was interrupted before its saved settings were cleared. Your previous preferences may still be active."	34
Cited	"E-2194 means the Ashby tumble dryer's factory reset was interrupted before its saved settings could be cleared, so your previous preferences may still be active. [P1]"	40

output tokens

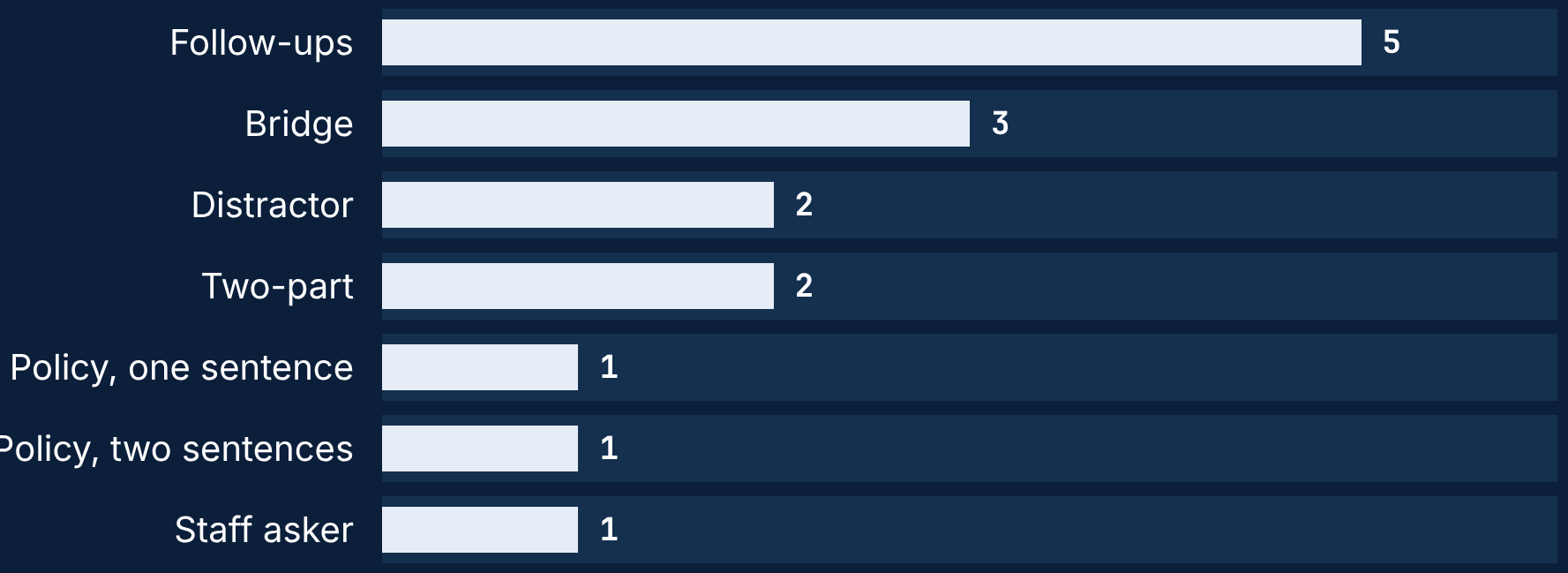
The same answer, with a pointer

Same facts, near-identical words, and a marker that E10's check can verify.

Finding

The flips were scattered, not a pattern

The 15 answerable questions that flipped either way, by kind of question.



Read across

Most flips are follow-ups and bridge questions, the kinds search finds hardest (E6, E7). Any small change tips them either way.

What this does not prove

- **One model.** A model that writes long answers, or already cites unasked, may react differently.
- **"Only" changed too.** The cited instruction also says "using only the passages".
- **Right by the judge.** Answers were marked by gemini-3.6-flash, not checked by hand for these runs.

For engineers

Run it yourself

The whole field note is one script over two committed runs.

```
# compare E9's and E10's runs: no key, no cost
$ uv run python field-notes/s2-e11-citation-rule/analyse.py
```

Same passages	analysis.json confirms every prompt held the same five passages in both runs.
Cost	Nothing. No model call: both runs were already paid for in E9 and E10.

For leaders

Require citations, and check what else changed

Here, citations cost two tokens per answer and changed nothing else. That is worth confirming before you mandate them.

Ask your team

Do our answers cite their sources? If not, what is stopping us?

Ask your team

When we change an instruction, do we compare answers question by question, or only the totals?

Ask your team

Could we answer this from runs we already have, before paying for new ones?

Decision rule

Asking for citations costs nothing in accuracy here: require them, and check them.

Both lessons hold	E9’s "not found" sentence and E10’s citations work together. Right answers and "not found" replies held; answers grew by 2 tokens.

Next: S2 E12, a field note

Swapping the embedding model. Is a bigger, pricier model an upgrade by default?

Where this fits

The series map

S1 LLMs	**S2** Retrieval	**S3** Strategy
S4 Adapting	**S5** Agents	**S6** Economics
S7 Data	**S8** Evaluation	**S9** AI teams
S10 Reliability	**S11** Security	**S12** Delivery
S13 Performance	**S14** Resilience	**S15** Governance

☐ Leadership season

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Episode guide brendankellyai.github.io

Code github.com/BrendanKellyAI/applied-ai-lab